

Aegis Alpha

An autonomous, explainable SPY/QQQ options agent where AI ranks opportunities but deterministic Python controls every order.

PROBLEM AND ORIGINAL IDEA

AI-generated trading theses can sound confident while inputs are stale, liquidity is weak, or portfolio risk is already exhausted. Aegis Alpha separates **judgment** from **authority**: an optional constrained ranker may choose among valid candidates, but it cannot weaken, skip, or rewrite a risk gate. Any missing data, malformed schema, contradictory signal, API error, or reconciliation mismatch becomes **NO_TRADE**.

FIXED STRATEGY

Five-minute SPY and QQQ bars feed EMA trend, VWAP position, RSI, and realized-volatility features. The regime is bullish, bearish, or neutral. The candidate layer emits only same-expiry bull-call or bear-put debit spreads with 7-21 DTE and a long-leg absolute delta near 0.35-0.55. The executor submits one atomic two-leg limit order with a deterministic `client_order_id`.

AGENT CYCLE

Market Data -> Regime -> Candidate -> Critic -> Risk Guard -> Executor -> Monitor -> Audit Log. The critic rejects incomplete contracts, unsupported structures, weak liquidity, contradictory signals, and existing exposure. The monitor reprices open spreads and closes at +40%, -30%, signal invalidation, or the 15:45 ET risk cutoff.

INFRASTRUCTURE

alpaca-py is the primary typed Trading and Market Data client. SQLite stores decisions, raw errors, orders, fills, snapshots, locks, and reconciliation evidence. The Alpaca CLI is isolated behind a JSON adapter and independently checks account, orders, and positions. Any SDK/CLI disagreement halts new orders. Alpaca MCP is configured only for judge-facing inspection; it is outside the execution path.

RISK GATE	FIXED LIMIT
Per trade	min(\$500, 0.5% equity)
Open risk	\$1,500 across max 3 positions
Frequency	1 new trade / underlying / day
Cooldown	30 minutes after order or exit
Quotes	<=60 sec old; width <=15% midpoint
Daily halt	1.5% equity drawdown
Endpoint	Paper only; live settings rejected
Kill switch	Blocks every new order

VERIFICATION STATUS

Locally verified: 28 tests passed, 5 synthetic market regimes replayed, 72% measured code coverage, dry-run submission lock, idempotency, forced exits, malformed-AI failure, sanitized export, and multi-leg request construction. **Credential-gated:** one paper integration test is skipped until Alpaca credentials exist; no filled trade or paper P&L is claimed.

DEMO AND DISCLOSURE

The read-only Streamlit dashboard shows equity, P&L, positions, the latest thesis, critic result, risk checks, order timeline, and reconciliation status from sanitized snapshots. This is a paper-trading prototype for demonstration and research. It is not financial advice and does not establish real-market profitability.